TRASNA: Technical Appendix

This appendix provides the rigorous evidence base, financial modelling, and technical assumptions underpinning the TRASNA proposal. TRASNA is designed to serve **all residents of County Dublin** — across Dublin City, Fingal, South Dublin, and Dún Laoghaire-Rathdown — from the first day of operation. This document is designed to withstand detailed scrutiny by the Department of Transport, the Department of Public Expenditure, and the National Transport Authority.

1. Fleet Sizing and Peak Demand Modelling

The core operational claim of TRASNA is that a fleet of 51,000 vehicles can guarantee a maximum wait time of 3 minutes across all of County Dublin (Dublin City, Fingal, South Dublin, and Dún Laoghaire-Rathdown), even at peak hours. This is derived from a combination of throughput modelling, spatial queuing theory, and the integration of bus lane access.

1.1 Total Addressable Demand

According to Census 2022, the population of Dublin city and county is 1,458,154 [1]. The NTA's National Household Travel Survey (2023) indicates that the average person in the Greater Dublin Area makes approximately 2.6 trips per day [2]. This yields a total of **3.79 million daily trips** across all modes.

We conservatively estimate that **65%** of all trips (single commuters, couples, errands, social trips) are addressable by a two-seat vehicle, excluding heavy goods, school buses, and multi-person family trips. Assuming TRASNA captures **40%** of this addressable market at full scale, the system will handle approximately **985,000 trips per day**.

1.2 The Design Peak

The PM peak (17:00–19:00) is the busiest period for ride-hailing. The worst single hour accounts for approximately 10.5% of daily trips, giving a **Design Peak of 118,285 trips per hour** (nearly 2,000 trips per minute).

1.3 Vehicle Throughput and Bus Lane Access

As a licensed public transport service, TRASNA vehicles will have statutory access to Dublin's 220km network of bus lanes. This significantly reduces journey times on arterial routes during peak hours.

The average in-vehicle trip duration in the Dublin urban area without bus lane access is approximately 18 minutes [2]. With bus lane access, we estimate the average peak journey time falls to 14 minutes. Adding 2.5 minutes for repositioning gives a total cycle time of 16.5 minutes. Therefore, each vehicle can complete **3.64 trips per hour**.

To serve 118,285 trips per hour at a 70% utilisation rate (the industry benchmark for high-quality service), the system requires **46,422 vehicles**.

1.4 Spatial Coverage

At an average urban repositioning speed of 25 km/h, a vehicle covers 1.25 km in 3 minutes, creating a coverage circle of 4.9 km^2. To cover Dublin's effective populated service area (~600 km^2), a minimum of **122 idle vehicles** must be evenly distributed. The binding constraint is therefore throughput, not spatial coverage.

Adding a 10% buffer for charging and maintenance to the throughput requirement yields the final recommended fleet size of **51,000 vehicles**.

1.5 Stress Test: Major Events (Croke Park)

The most extreme single-point demand spike in Dublin occurs after an All-Ireland Final at Croke Park (capacity 82,300). If 30% of attendees (25,000 people) request a vehicle within a 20-minute window, the local demand reaches 1,250 trips per minute.

The system manages this through four mechanisms:

1. **Pre-positioning:** The central AI detects the scheduled event and pre-positions thousands of idle vehicles in surrounding zones (Drumcondra, Ballybough) during the second half of the match.
2. **Geofenced Hubs:** Passengers are directed to designated pickup zones rather than individual street corners, mirroring current coach departure management.
3. **Capacity Headroom:** A Sunday afternoon match occurs during off-peak background demand (approx. 25–30% fleet utilisation), leaving over 40,000 vehicles available to absorb the surge.
4. **Multi-Modal Integration:** TRASNA does not replace mass transit; it complements it. The DART and Luas carry the bulk of the crowd, with TRASNA providing the "last mile" connection from the station to the home.

While the 3-minute wait guarantee applies to normal operations, wait times immediately outside the stadium post-match are modelled at 8–12 minutes—a significant improvement over current taxi queuing times.

2. Capital Cost and Financing

2.1 Vehicle Unit Cost

The TRASNA vehicle is a purpose-built, two-seat electric microcar. It does not require the performance, range, or luxury amenities of a retail passenger car.

- **Base Vehicle:** The Citroën Ami retails at approximately €7,990 [3]. A bespoke fleet vehicle manufactured at scale (64,000 units) can be produced for €8,000–€12,000.
- **AV Sensor Suite:** Waymo's 6th-generation driver system costs less than $20,000 (€18,500) [4]. A simpler system designed for low-speed urban environments will cost significantly less. We estimate the sensor suite at €8,000–€12,000.
- **Total Unit Cost:** €16,000–€24,000. We use a central estimate of **€20,000 per vehicle**.

2.2 Total Capital Expenditure (CapEx)

- **Fleet (51,000 units @ €20,000):** €1.02 billion
- **Charging Infrastructure & Depots:** €150 million

- **Software & Control Centre:** €50 million
- **Total CapEx: €1.22 billion**

This compares highly favourably to the MetroLink central cost estimate of €9.5 billion to €12.25 billion [5].

2.3 Financing Options

The €1.22 billion capital requirement can be financed through several mechanisms:

1. **Direct Exchequer Funding:** Funded from the capital envelope of the National Development Plan.
2. **Local Authority Green Bond:** A 20-year bond issued by the four Dublin local authorities, serviced by system revenues. At a 3.5% yield, annual debt service would be approximately €104 million.
3. **Public-Private Partnership (PPP):** A private consortium finances the fleet and infrastructure in exchange for a fixed availability payment, with the State retaining ownership of the brand, data, and fare revenue.

3. Operating Economics

3.1 Operating Expenditure (OpEx)

Industry consensus projects that autonomous robotaxi operating costs will fall to 0.30–0.50 per mile by 2030 [6]. For a lightweight microcar, energy and maintenance costs are even lower.

- **Estimated OpEx per km:** €0.25
- **Average Trip Distance:** 6 km
- **OpEx per trip:** €1.50

3.3 Operating Economics and Revenue Model

TRASNA operates on a subscription model with no per-journey fares. The pricing tiers are designed to be universally affordable while generating substantial revenue. Based on Dublin's population demographics (Census 2022), the revenue model projects a 40% overall take-up rate at maturity:

- **€365/year:** Annual adult subscription (approx. 182,000 subscribers)
- **€100/year:** Concession for students, under-18s, pensioners, and disability benefit recipients (approx. 167,000 subscribers)
- **€50/month:** Rolling monthly subscription for all users (approx. 145,000 subscribers)
- **€25/week:** Rolling weekly subscription for all users (approx. 87,000 subscribers)

This tiered structure generates approximately **€285 million in annual subscription revenue**.

In addition, TRASNA will generate commercial revenue through vehicle exterior sponsorship and non-intrusive in-app advertising, similar to the Dublin Bikes and Dublin Bus models. Given the fleet size and visibility, this is projected to generate an additional **€15–€25 million annually**, bringing total system revenue to approximately **€305 million per year**.

Annual Operating Costs (Base Case: 51,000 vehicles)

- **Fleet maintenance:** €51m (€1,000/vehicle)

- **Electricity:** €12m (8,000km/year × 0.15kWh/km × €0.20/kWh)
- **Charging infrastructure maintenance:** €15m
- **Staff (1,500 employees):** €83m (Average fully-loaded cost €55,000 p.a.)
- **Insurance & liability:** €25m
- **Software licensing / R&D / OTA:** €30m
- **Vehicle replacement reserve:** €146m (7,300 vehicles/year × €20,000)
- **Central operations, admin, marketing:** €20m
- **Contingency (10%):** €38m
- **Total Annual Operating Cost: €420m**

The shortfall between the €420m operating cost and the €305m revenue is approximately **€115 million per year**. This represents the Public Service Obligation (PSO) subvention required to run the system. This is highly efficient compared to existing public transport funding models; in 2023, the PSO payment for Dublin Bus, Bus Éireann, and Iarnród Éireann was over €500 million [7].

3.4 The Best-Case Scenario: 80% Take-Up

The base model assumes a conservative 40% take-up rate. However, if TRASNA becomes as ubiquitous as the smartphone, reaching 80% take-up across Dublin, the system scales efficiently while transforming the city.

At 80% take-up (1.16 million subscribers), TRASNA would capture 70% of all short-to-medium journeys in Dublin. To meet this peak demand, the fleet would need to expand to **123,000 vehicles**.

- **Capital Cost:** €2.96 billion (still a fraction of MetroLink's central estimate)
- **Annual Revenue:** €681 million
- **Annual OpEx:** €1.45 billion
- **PSO Subvention:** €772 million per year

Crucially, at this scale, TRASNA would remove an estimated **398,000 private cars** from Dublin's roads, effectively ending car dependency across the region and making Dublin the world's first post-car county.

4. Operations, Maintenance, and Security

4.1 Fleet Lifespan and Maintenance

A TRASNA microcar is designed for a 7-year or 250,000km lifespan in heavy fleet use. Electric vehicles have significantly lower maintenance requirements than internal combustion engine (ICE) vehicles, with savings estimated at up to 50% [10]. Maintenance is handled centrally at dedicated TRASNA depots located strategically across the four local authority areas. Vehicles are programmed to autonomously return to a depot when their sensors detect a maintenance need, a drop in battery health, or when they require deep cleaning.

4.2 Staffing and Employment

While the vehicles are driverless, the system requires a substantial workforce to operate across the county. Based on industry benchmarks from Waymo (which operates approximately 40 vehicles per remote assistance operator

[11]), a 51,000-vehicle fleet will require a dedicated workforce of approximately 1,500 direct employees. This includes:

- **Remote Assistance Operators:** Handling edge cases and providing routing guidance when a vehicle encounters an unmapped obstacle.
- **Depot Technicians:** Mechanics, electricians, and cleaning staff.
- **Software Engineers and Data Scientists:** Managing the central AI, routing algorithms, and app infrastructure.

4.3 Cybersecurity, Critical Infrastructure, and Data Privacy

TRASNA operates as critical national infrastructure. The cybersecurity architecture is built on three principles:

1. **On-Board Autonomy:** Vehicles do not rely on a continuous internet connection to drive safely. All collision avoidance, sensor fusion, and immediate navigation processing happen on the vehicle's internal computer. The central network only provides high-level routing instructions.
2. **Network Isolation:** The fleet operates on a secure, encrypted sub-network, physically and logically separated from public internet access.
3. **Over-The-Air (OTA) Security:** Software updates are delivered via secure OTA channels, allowing the rapid deployment of security patches across the entire fleet simultaneously [12].

Data Privacy Commitment: Because TRASNA handles the daily travel patterns of hundreds of thousands of citizens, privacy is written into its charter. TRASNA will **never** sell personal data, location history, or travel patterns to third parties. It operates strictly as a public service, not a data broker.

4.4 Continuous Improvement Programme

TRASNA is not a static piece of infrastructure. Like modern consumer electronics, the system improves continuously. Every year, the fleet receives major OTA software upgrades—delivering smoother cornering, better route prediction, and new app features. Hardware improvements (such as enhanced seating, better sound systems, or upgraded sensors) are rolled out continuously as older vehicles reach the end of their 7-year lifespan and are replaced with newer models.

5. Urban Transformation and V2V Technology

5.1 Autonomous Intersection Management (AIM)

TRASNA vehicles will utilise Vehicle-to-Vehicle (V2V) communication to negotiate intersections without stopping. This "zipper merging" eliminates the need for traffic lights within the closed system. Peer-reviewed research demonstrates that AIM can increase intersection capacity by over 40% and virtually eliminate stop-start delays [8].

5.2 The Canal Cordon

In 2024, 38,390 private cars crossed the canal cordon into central Dublin during the AM peak [9]. TRASNA enables the complete restriction of private through-traffic within the canals, reclaiming massive amounts of urban space for pedestrians, cyclists, and public realm improvements, while still providing door-to-door access for those who need it.

6. Social Equity

TRASNA is designed as universal public infrastructure. It provides independent mobility for the elderly, the disabled, and those who cannot drive or afford a car.

An estimated 277,000 people in Dublin county are recipients of the Free Travel Scheme [1]. These users will have their €1/day subscription covered by the Department of Social Protection, ensuring that the transition to autonomous transport leaves no one behind.

7. Implementation Timeline and Phased Rollout

A geographically constrained pilot (e.g., limiting operations to a single local authority area) would fail to demonstrate the system's core value proposition: universal, door-to-door mobility. Therefore, TRASNA will be deployed across the entire county from Day 1, but access will be controlled via a phased, limited-user beta trial.

- **2026:** Government approval, establishment of statutory body, and procurement.
- **2027:** Delivery of initial 5,000 vehicles. Construction of county-wide charging infrastructure.
- **2028 (County-Wide Beta):** Fleet expands to 15,000 vehicles. Access is granted to 50,000 beta users (selected to represent a cross-section of demographics and geographies) to stress-test the full door-to-door experience at scale.
- **2029 (Full Launch):** Fleet reaches 51,000 vehicles. System opens to all residents across the four Dublin local authorities. Canal cordon restrictions take effect in the city centre, alongside suburban traffic calming measures.

8. A World First: The Lemass-Whitaker Legacy

TRASNA is more than a transport project; it is a statement of national ambition. In the late 1950s and 1960s, Seán Lemass and T.K. Whitaker transformed Ireland from an agrarian, protectionist economy into a modern, outward-looking industrial state. Today, Ireland is the European home of the world's leading technology and AI companies, yet our physical infrastructure often lags behind our digital economy.

By building the world's first publicly owned, city-wide autonomous transport network, Ireland would cement its position as a global leader in applied AI and smart infrastructure. TRASNA would generate massive, sustained positive publicity worldwide. It would serve as a powerful, tangible symbol that the State is maintaining and renewing the forward-looking, innovative industrial strategy adopted by Lemass and Whitaker, proving that Ireland is not just a host for global technology, but a pioneer in deploying it for the public good.

References

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